# BIHAR 2026

Systeme de Prévision Météorologique & Classification Multimodales

Yamoussoukro, Cote d'Ivoire | 6.82 N, 5.28 O | Tropical subequatorial


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Master MLOps — ESTIA | Avril 2026

# Plan de la Présentation


01 METEO

## Donnees & EDA

Collecte - analyse exploratoire - feature engineering

02 MODELE

## Modelisation Meteo

Prophet vs TFT vs LSTM — architectures detaillees — prevision 72h

03 MULTI

## Classification Multimodale

CNN / EfficientNet — MLP / BiLSTM — Fusion early+joint

04 MLOPS

## MLOps / DevOps

Docker — Airflow — CI/CD GitHub Actions — FastAPI — Monitoring

05 DEMO

## Demonstration

Interface web — Dashboard Grafana — Tests automatises (16 tests)

# Architecture Globale — Vue d'ensemble du Systeme

## SOURCES

**Open-Meteo API**
temp 3h / 72h

**Dataset COCO**
images multi-label

**Dataset SNLI**
captions texte

## STOCKAGE

**SQLite weather.db**
predictions + meteo

**MLflow Registry**
modeles + metriques

**Volume Docker**
persistence

## TRAITEMENT

**LSTM PyTorch**
prevision 72h

**EfficientNetB0**
classif image

**BiLSTM + MLP**
classif texte

## EXPOSITION

**FastAPI :8000**
REST endpoints

**Flask Web :5000**
interface utilisateur

**Grafana :3000**
dashboards meteo

# Données & Analyse Exploratoire

Yamoussoukro · Open-Météo API · Feature Engineering 49 vars

# Yamoussoukro — Capitale Politique de la Cote d'Ivoire

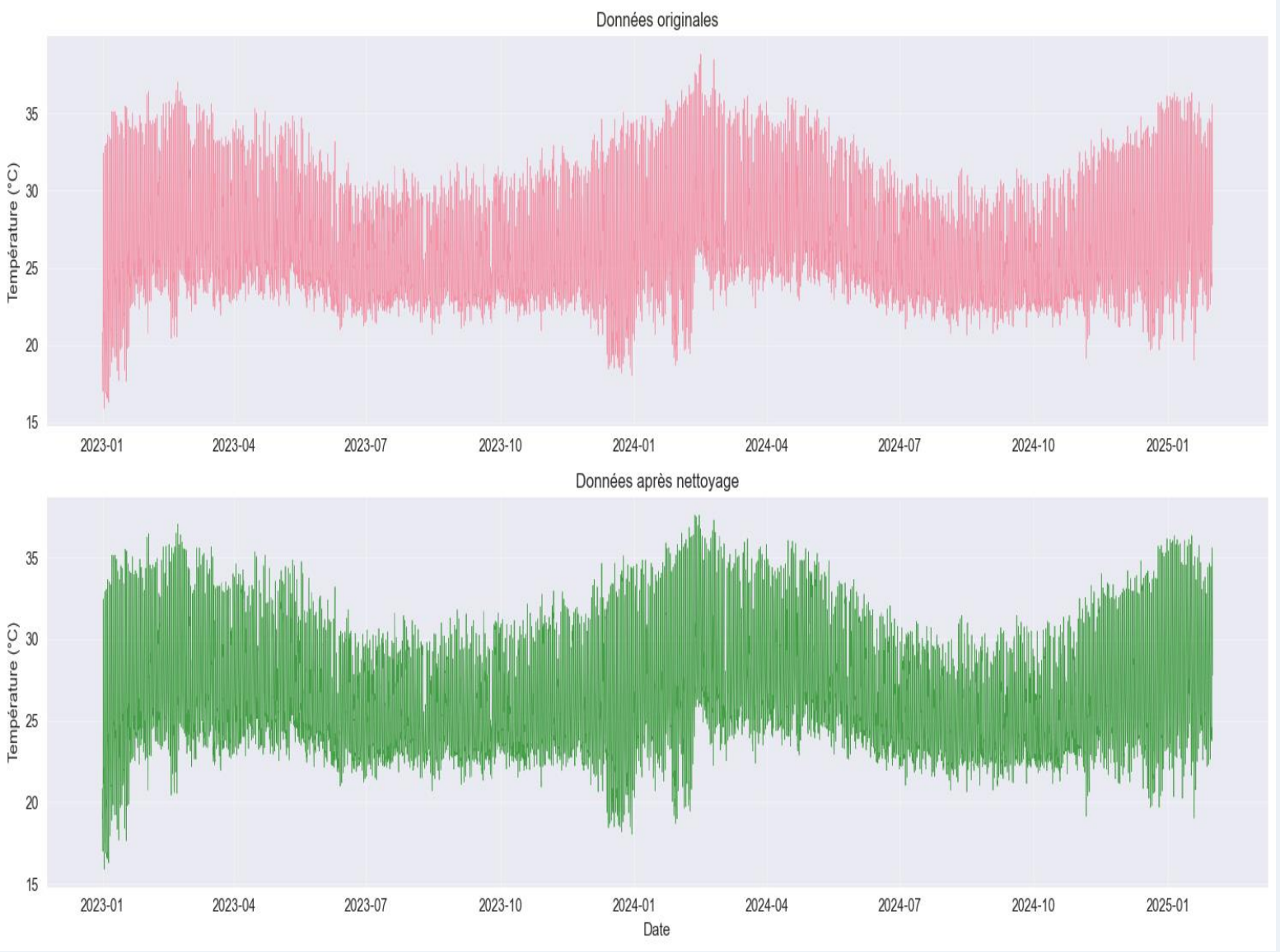
## Localisation & Climat

| | |
|---|---|
| **Pays** | Cote d'Ivoire (Afrique de l'Ouest) |
| **Coordonnees** | 6.816667 N \| -5.283333 E |
| **Altitude** | 213 m |
| **Climat** | Tropical subequatorial (Aw) |
| **Temp. moy. annuelle** | 26.5 degres C |
| **Pluviometrie** | 1 200 mm/an (2 saisons) |
| **Saison seche** | Novembre → Mars |
| **Saison pluies** | Avril → Octobre |
| **Caracteristique** | Tres peu de variation saisonniere |

## Donnees Open-Meteo API

| | |
|---|---|
| **Variable cible** | temperature_2m (degres C) |
| **Frequence** | Toutes les 3 heures (8 pts/jour) |
| **Periode** | 2023-01-01 → 2025-01-31 |
| **Nb points total** | ~18 288 observations |
| **Split train** | 70% (2023-01 → 2024-07) |
| **Split validation** | 15% (2024-08 → 2024-11) |
| **Split test** | 15% (2024-12 → 2025-03) |
| **Horizon prevu** | 24 pas x 3h = 72h (3 jours) |
| **Historique modele** | 24 pas = 3 jours d'historique |

# Analyse Exploratoire




## Data Cleaning Summary

### Comparaison

- count : 18288 → 18288
- mean : 26.52 → 26.53
- std : 3.70 → 3.70
- min : 15.90 → 15.90
- median: 25.60 → 25.65
- max : 38.85 → 37.60

### Méthodes

- Z-score > 3 → 11 outliers détectés
- ΔT > 5°C/h → 59 sauts extrêmes
- Interpolation linéaire appliquée
- 70 valeurs corrigées (0.38%)

### Top 5 sauts de température

10.10°C 9.85°C 9.25°C 9.20°C 8.70°C

# Feature Engineering — 49 Features Temporelles

| CYCLIQUES (calendrier) | LAG FEATURES (valeurs passees) | ROLLING STATS (fenetres glissantes) | DIFFERENCES (variations) | SEQUENCEMENT (entree LSTM) |
|---|---|---|---|---|
| *sin/cos(2pi x t/T)* | *temp(t - k*3h)* | *mean(temp[t-w:t])* | *temp(t) - temp(t-k)* | *(batch, 24, 49)* |
| hour_sin / hour_cos | lag_1 … lag_24 | rolling_mean 3,6,8,12,24 | diff_1 = temp(t) - temp(t-1) | Fenetre : 24 pas |
| day_sin / day_cos | lag_36, lag_48 | rolling_std 3,6,8,12,24 | diff_2 = temp(t) - temp(t-2) | = 3 jours d'historique |
| month_sin / month_cos | = temp. t-1h, t-3h ... | = tendance locale | diff_8 = ecart 24h | Shape : (24, N_feat) |
| hour, dayofweek | t-8h, t-24h, t-48h | = volatilite locale | = tendance instantanee | N_feat = 49 features |
| month, is_weekend | | | | |

TOTAL : 49 features | Entree LSTM : shape (batch, 24, 49) | Normalisation Z-score sur features ET cible | StandardScaler.fit() sur train uniquement

# Modelisation Météorologique

Prophet · TFT (Temporal Fusion Transformer) · LSTM 3 couches

# Trois Approches de Prevision — Comparaison Architecturale

| Critere | Prophet | TFT | LSTM (prod) |
|---|---|---|---|
| Type | Statistique additif | Transformer hybride | Deep Learning seq. |
| Multi-step natif | Oui | Oui | Adapte |
| Features ext. | Limitees | Oui (VSN) | Oui (49 features) |
| Attention | Non | Multi-head (4 tetes) | Non |
| Intervalles conf. | Oui (incertitude) | Oui (quantiles) | Non |
| Parametres | ~Quelques params | ~100K params | ~50K params |
| MAE test | ~1.5 C | ~1.58 C | 0.052 C |
| MAE production | Non deploye | Non deploye | 0.807 C (56pts) |
| **STATUT** | **Reference** | **Experimentation** | **Production** |

LSTM selectionne en production : meilleures performances + pipeline full PyTorch integre dans l'architecture MLOps

# Prophet — Modele Additif Decomposable (Facebook, 2018)

***Formule : y(t) = g(t) + s(t) + h(t) + epsilon***

| g(t) Tendance | s(t) Saisonnalite | h(t) Jours feries | epsilon Residu |
|---|---|---|---|
| Modele lineaire par morceaux | Serie de Fourier | Effets ponctuels | Bruit non explique |
| OU modele logistique | Saison annuelle (annee) | Fetes nationales CI | Hypothese : i.i.d. |
| Changepoints automatiques | Saison hebdo (7 jours) | Evenements specifiques | Pas capte par modele |
| Flexibilite reglable | Saison journaliere (24h) | Ajustement manuel | -> Intervalles de conf. |

**Resultats Prophet (temperature Yamoussoukro) :**

**MAE: ~1.5 C**

**RMSE: ~2.1 C**

**Statut: Reference baseline**

# TFT — Temporal Fusion Transformer (Lim et al., 2021)

| INPUTS | Variable Selection Networks | LSTM Encoder / Decoder | Multi-Head Attention | Gated Residual Networks | OUTPUTS Quantiles |
|---|---|---|---|---|---|
| Features connues futur | Selectionne features | Encoder: historique 24p | attention_heads = 4 | Flux info controle | 5 quantiles: |
| (calendrier: hour, day...) | pertinentes auto. | Decoder: horizon 24p | Dependances long-terme | Gate = sigmoid() | 10%, 25%, 50%, |
| Features passees observees | Poids d'importance | hidden_size = 64 | Pondere tous les pas | Residual connection | 75%, 90% |
| (lag, rolling, diff) | par variable | lstm_layers = 2 | Interpretable | Stabilise gradient | Intervalles de conf. |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| hidden_size=64 | lstm_layers=2 | attention_head_size=4 | dropout=0.1 | hidden_continuous_size=16 | loss=QuantileLoss | lr=1e-3 | batch=64 | max_epochs=60 | patience=10 |

**TFT — Resultats :**

| MAE test: ~1.58 C | RMSE test: ~2.1 C | Quantiles: 10%~90% | Statut: Experimentation | Avantage: Interpretabilite |
|---|---|---|---|---|

*Librairie: pytorch-forecasting + pytorch-lightning | Ref: Lim et al. (2021) — NeurIPS*

# LSTM Multi-couches — Architecture Detaillee (PyTorch)

| INPUT (24, 49) | LSTM 1 128 units Dropout 20% | LSTM 2 64 units Dropout 20% | LSTM 3 32 units Dropout 20% | Dense 64 + ReLU | Dense 32 + ReLU | OUTPUT 24 vals (72h) |
|---|---|---|---|---|---|---|

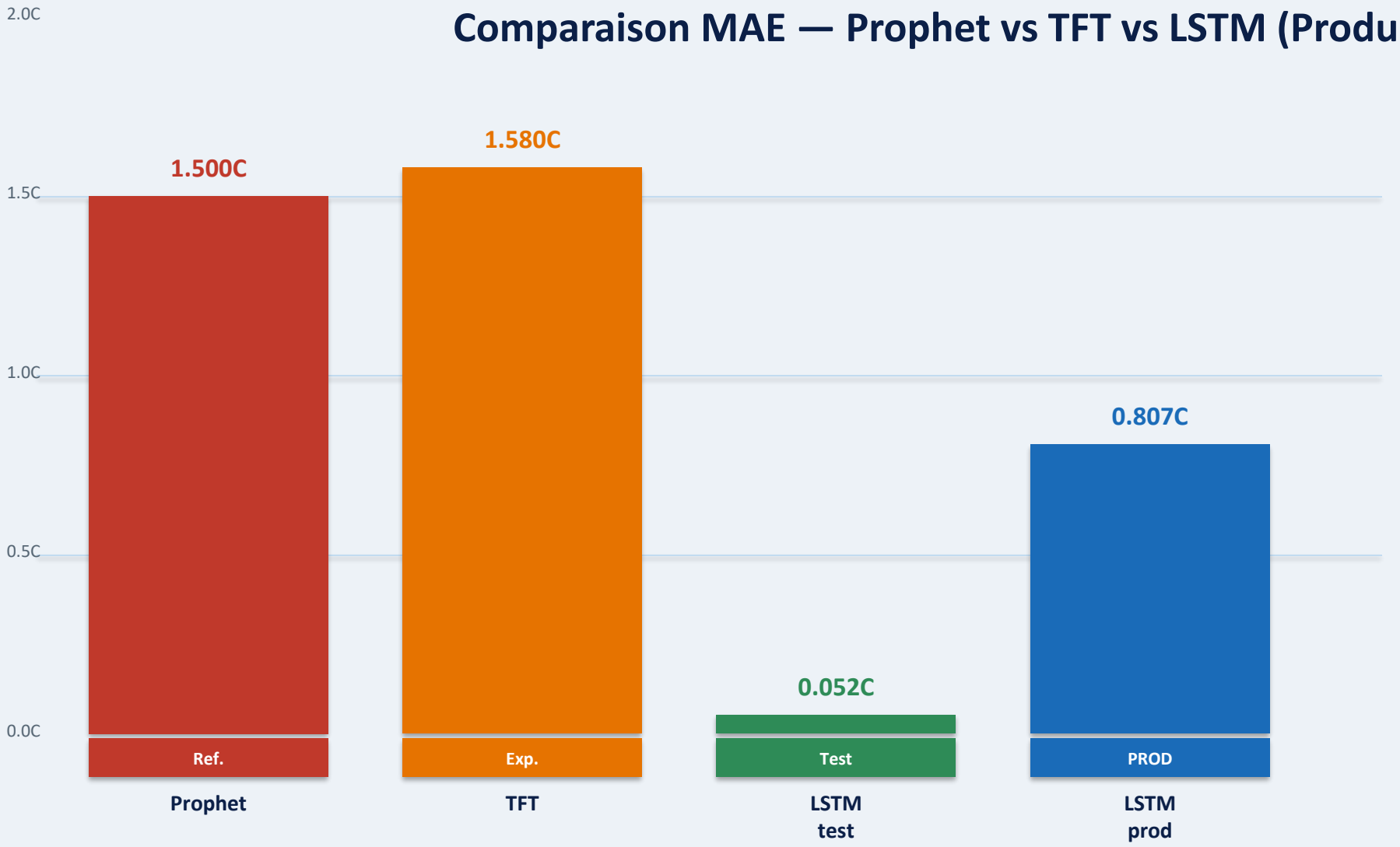
## Detail cellule LSTM :

| | | |
|---|---|---|
| Forget gate | *f(t) = sigmoid(Wf.[h,x] + bf)* | Oublie info passee |
| Input gate | *i(t) = sigmoid(Wi.[h,x] + bi)* | Quelle info stocker |
| Cell state | *C(t) = f*C(t-1) + i*tanh(Wc.[h,x])* | MAJ memoire |
| Output gate | *o(t) = sigmoid(Wo.[h,x] + bo)* | Sortie filtree |
| Hidden state | *h(t) = o(t) * tanh(C(t))* | Etat cache courant |

| Configuration | |
|---|---|
| Optimizer | Adam |
| Loss | MSELoss |
| lr | 1e-3 |
| Batch | 32 |
| Epochs | 100 max |
| Patience | 10 |
| Device | CPU/CUDA |
| Norm. | Z-score |

**MAE test=0.052 C | RMSE test=0.068 C | MAE prod=0.807 C | RMSE prod=1.082 C | 56 points evalues**

# Comparaison MAE — Prophet vs TFT vs LSTM (Production)




| MAE LSTM / horizon | |
|---|---|
| +3h (pas 1) | 0.034 C |
| +24h (J+1) | 0.051 C |
| +48h (J+2) | 0.058 C |
| +72h (J+3) | 0.069 C |
| Global | 0.052 C |

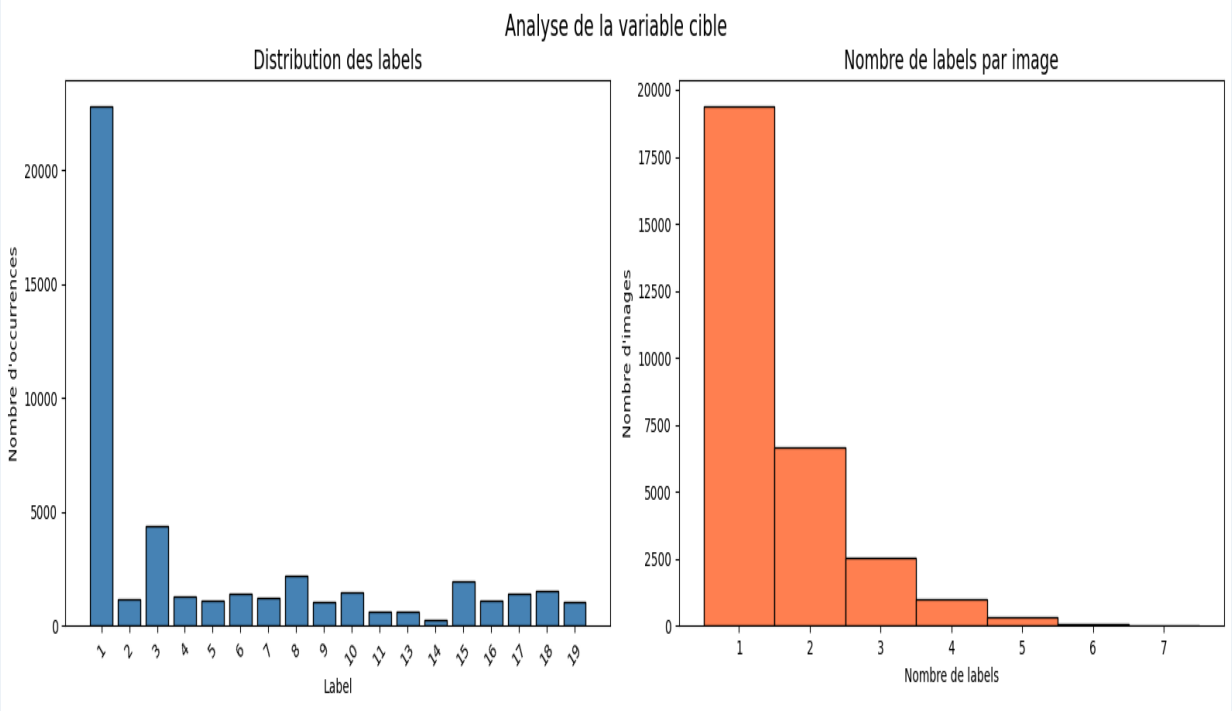
LSTM retenu en PRODUCTION : MAE 0.807 C sur donnees reelles | Biais : +0.61 C (surestimation legere)

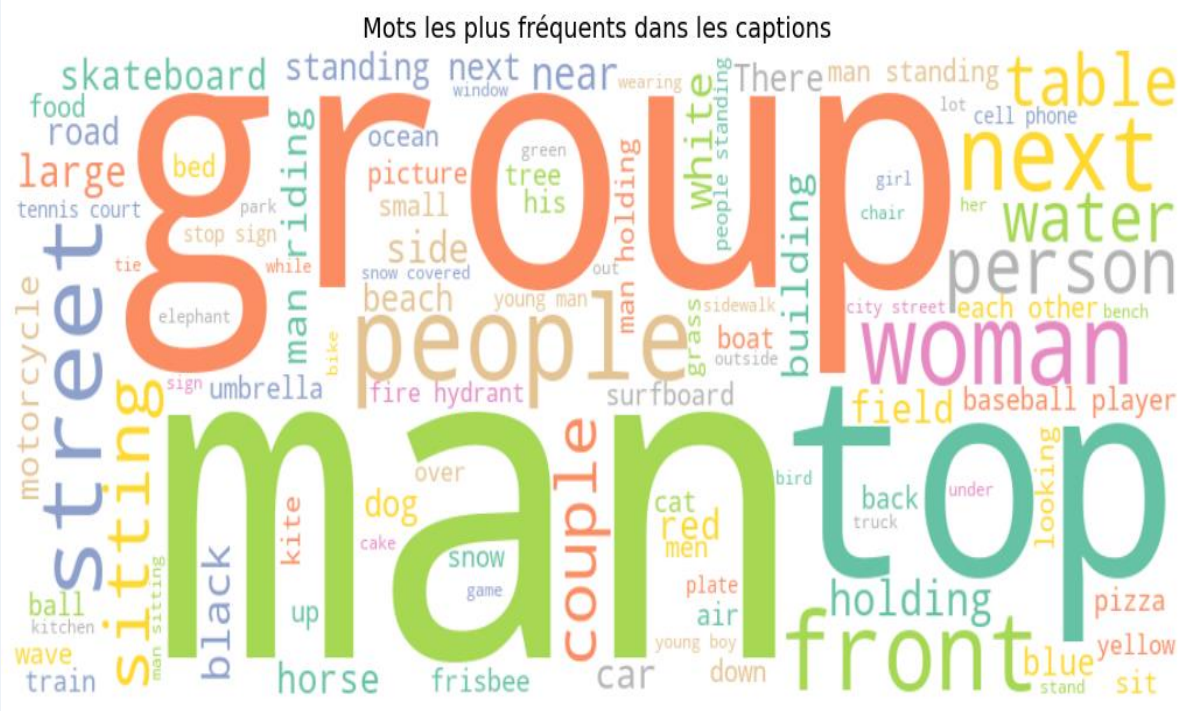
# Classification Multimodale

Images COCO · Textes SNLI · Fusion Early & Joint

# Analyse Exploratoire — Dataset Multi-Label (COCO/SNLI)

| | | | |
|---|---|---|---|
| **Classes** | 18 labels (1-19, sauf 12) | **Type** | Multi-label (N classes/image) |
| **Images** | Dataset Kaggle Multi-label 2023 40 000 images | **Captions** | Descriptions textuelles (SNLI) |
| **Train/Val/Test** | 70% / 15% / 15%: Train: 20997 \| Val: 4499 \| Test: 4500y_train shape: (20997, 18) | **Metrique** | F1 micro + F1 macro |




Distribution labels (image)

Vocabulaire des captions (texte)

# Classification d'Images — Pipeline & Architectures CNN

## Pipeline de Traitement des Images

## Trois Architectures Comparees

**Experimentation**

**Baseline CNN**

- 3 blocs Conv2D + MaxPool
- Filtres: 32 -> 64 -> 128
- BatchNorm apres Conv
- Dense 256 + Dropout 0.5
- Sigmoid (18 classes)
- Params: ~500K

**Transfer Learning**

**EfficientNetB0 Feature Extractor**

- EfficientNetB0 gele (frozen) 'IMAGENET1K_V1'
- GlobalAveragePooling2D
- Dense 512 + Dropout 0.4
- Dense 256 + Dropout 0.3
- Sigmoid (18 classes)
- Params backbone: 4M

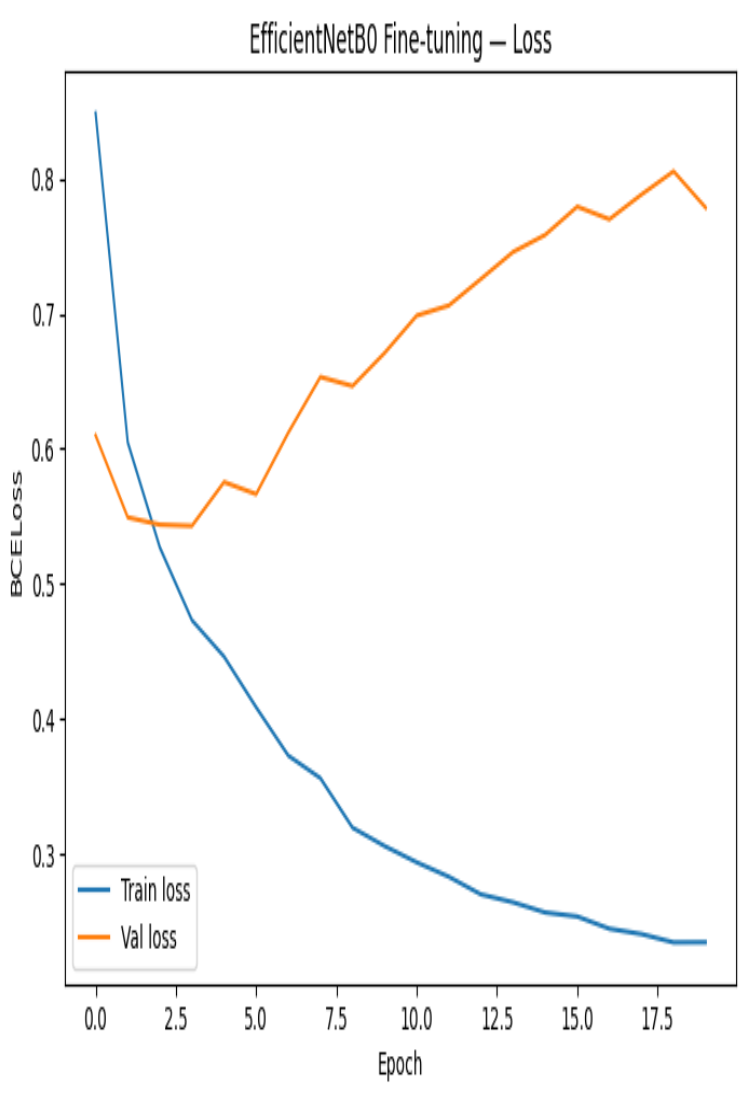
**PRODUCTION**

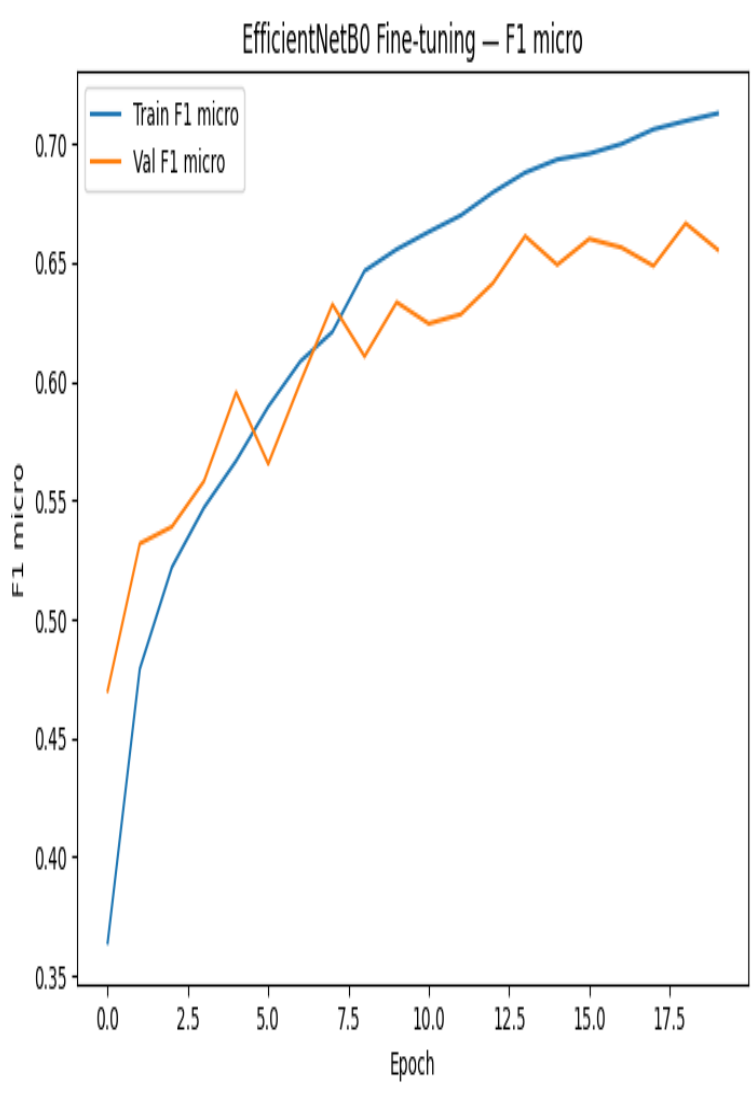
**EfficientNetB0 Fine-Tuning**

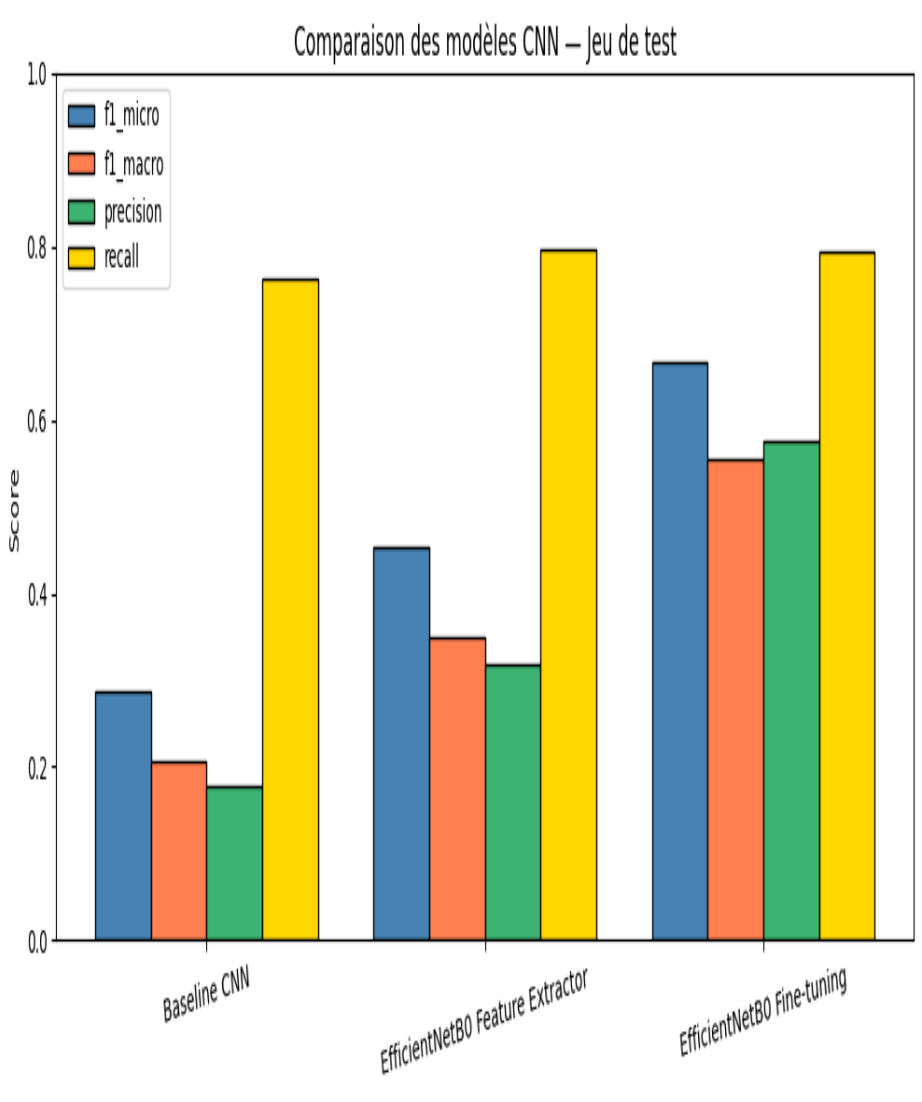
- Dernieres couches degelees
- Fine-tune bloc7 + tete
- lr tres faible: 1e-5
- Dense 512 + 256
- Sigmoid (18 classes)

*** MEILLEURE PERF.**

# Classification d'Images — Resultats & Courbes d'Apprentissage



# Classification de Texte — Pipeline & Trois Architectures NLP

## Pipeline commun de traitement (Captions)

Texte brut → lower + suppr punct. stopwords → TF-IDF (M1+M2) vecteur 5000d → Tokens (M3) indices vocab → Modele (LR / MLP / BiLSTM) → Sigmoid 18 sorties multi-label

## Trois Architectures Comparees

### M1 — TF-IDF + LogReg [RETENU]

| | |
|---|---|
| TF-IDF | max_features=5000<br>ngram_range=(1,2)<br>sublinear_tf=True |
| LogReg OvR | OneVsRestClassifier<br>LogisticRegression(C=1.0)<br>max_iter=1000 |
| Sortie | 18 classifieurs binaires independants |

**F1 micro = 0.8048 | Precision = 0.9217 [BEST]**

### M2 — TF-IDF + MLP

| | |
|---|---|
| Entree | TF-IDF (5000d) |
| Dense 512 | Linear + BN + ReLU<br>+ Dropout 0.4 |
| Dense 256 | Linear + BN + ReLU<br>+ Dropout 0.4 |
| Sortie | Linear(256,18)<br>+ Sigmoid |

**F1 micro = 0.7147 | Recall = 0.7752**

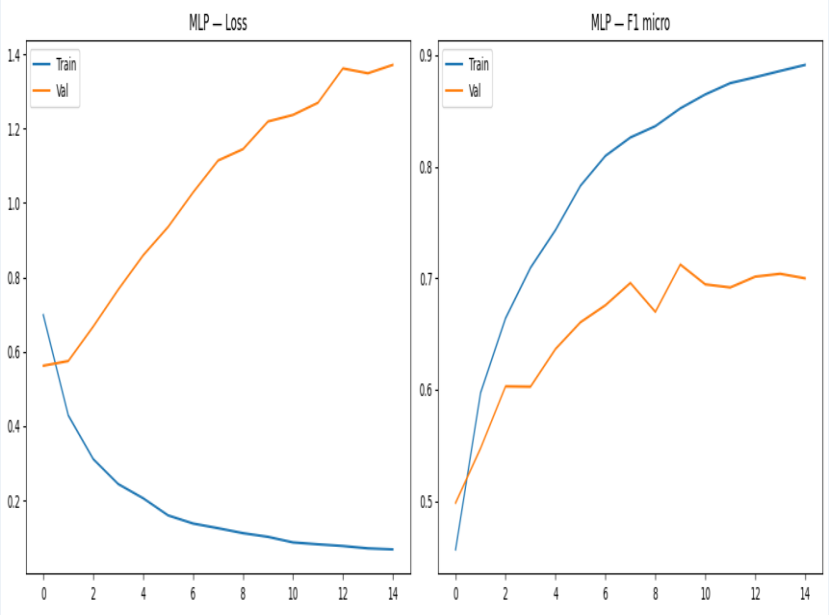
### M3 — Bi-LSTM (Deep Learning)

| | |
|---|---|
| Entree | Tokens indices<br>max_len=20, vocab=10002 |
| Embedding | Embed(10002, 128)<br>(appris, pas GloVe) |
| BiLSTM x2 | LSTM(128,128, bidir=True)<br>2 couches, dropout=0.3 |
| AvgPool | mean(out, dim=1)<br>-> (batch, 256d) |
| Dense 128 | Linear(256,128)+ReLU<br>+Dropout 0.3 |

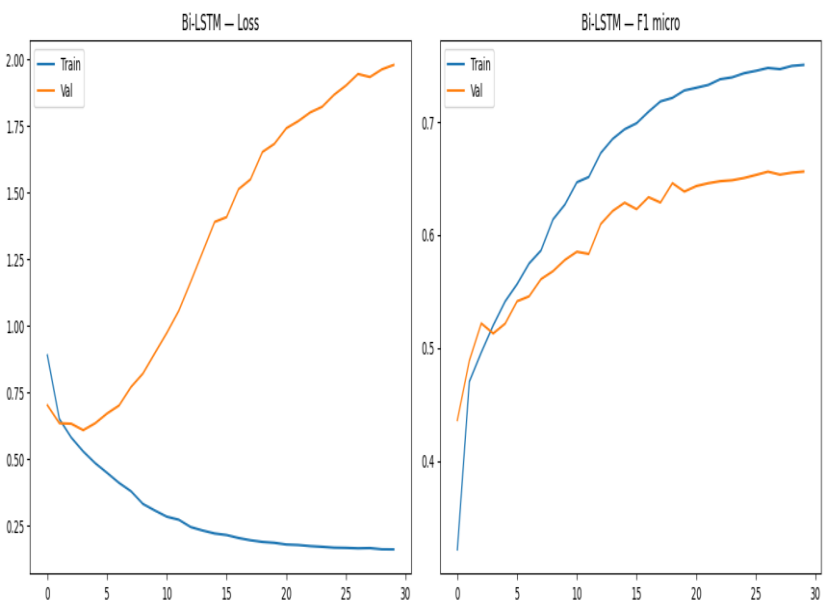
**F1 micro = 0.6575 | Recall = 0.7692**

**Modele retenu : TF-IDF + LogReg (F1=0.8048) — Le classifieur lineaire surpasse les approches DL sur ce corpus court | Precision = 0.9217**

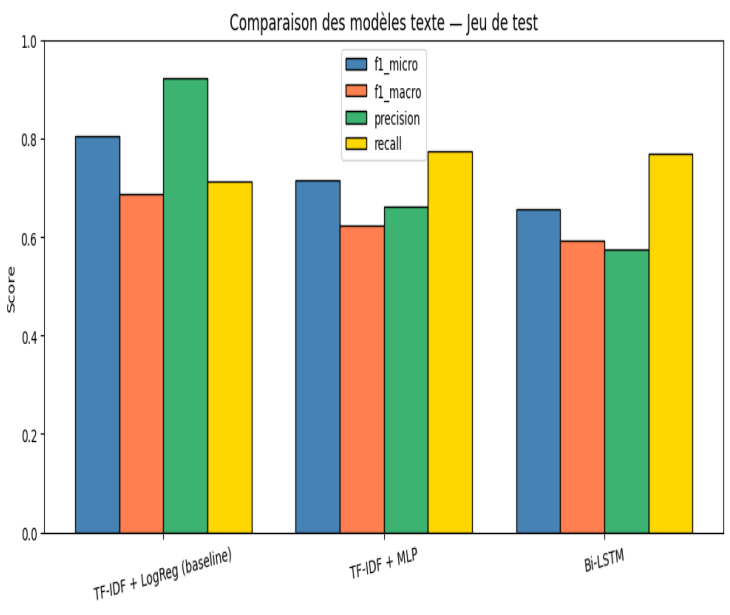
# Classification Texte — Resultats & Courbes d'Apprentissage




Courbes apprentissage MLP




Courbes apprentissage BiLSTM




Comparaison 3 modeles

| Modele | F1 micro | F1 macro | Precision | Recall | Statut |
|---|---|---|---|---|---|
| **TF-IDF + LR (baseline)** | **0.8048** | **0.6867** | **0.9217** | **0.7142** | **RETENU** |
| TF-IDF + MLP | 0.7147 | 0.6230 | 0.6629 | 0.7752 | Exp. |
| Bi-LSTM | 0.6575 | 0.5926 | 0.5741 | 0.7692 | Exp. |

**Modele retenu : TF-IDF + LogReg (F1=0.8048, Precision=0.9217) — Le classifieur lineaire surpasse MLP et BiLSTM sur les captions courtes**

# Fusion Multimodale — Architectures Early Fusion & Joint Fusion

## Early Fusion — F1 = 0.738 (MEILLEUR)

**Image (EfficientNet)**
features : 1280d

**Texte (BiLSTM+Attn)**
features : 256d

**Concatenation [1280 + 256 = 1536d]**

**Dense 512 + ReLU + BatchNorm + Dropout 0.4**

**Dense 256 + ReLU + Dropout 0.3**

**Linear(256, 18) + Sigmoid (18 classes multi-label)**

## Joint Fusion — F1 = 0.712

**Image branch**
EfficientNet -> Dense 512
-> features 512d

**Texte branch**
BiLSTM -> Dense 256
-> features 256d

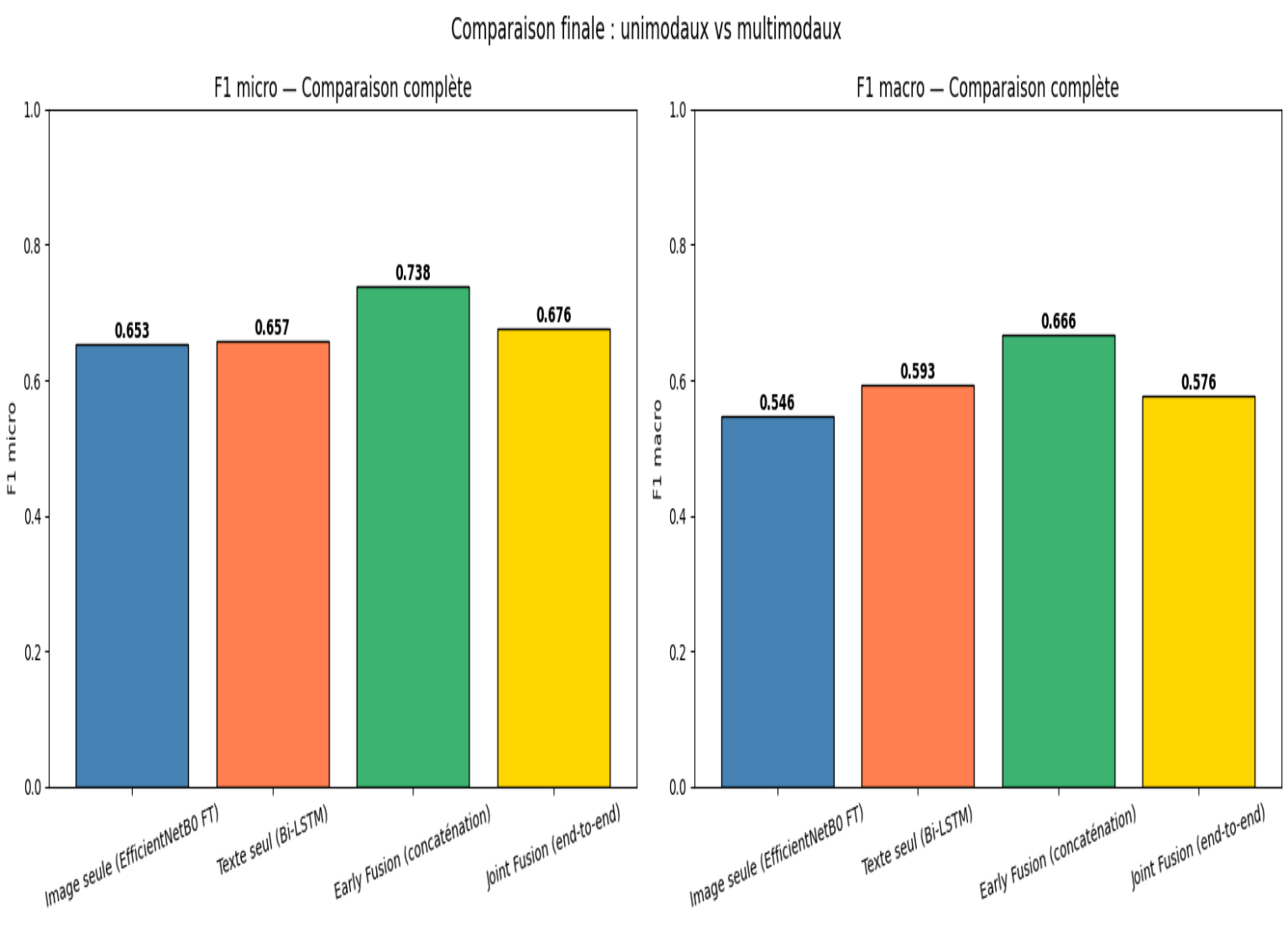
**Cross-Attention**
**Q = image | K = V = texte -> attention multimodale**

**Fusion MLP : Concat(attn, img, txt) -> Dense 512 -> Dense 256**

**Dense 256 + ReLU + Dropout 0.3**

**Linear(256, 18) + Sigmoid (18 classes)**

# Fusion Multimodale — Resultats & Analyse




## Synthese des Resultats

| | |
|---|---|
| **Early Fusion** | **F1 = 0.7383** |
| Joint Fusion | F1 = 0.712 |
| Image seule | F1 = 0.681 |
| Texte seul | F1 = 0.695 |
| Baseline LR | F1 = ~0.5 |

### Pourquoi Early Fusion > Joint ?

La concatenation directe avant les couches denses permet une fusion plus riche. Le cross-attention joint ajoute de la complexite sans gain.

**Conclusion : La fusion image+texte (Early Fusion) surpasse systematiquement les modalites separees — F1 +5.7% vs image seule**

# MLOps / DevOps

Docker · Airflow · CI/CD · FastAPI · Monitoring

# Architecture Docker Compose — 7 Microservices

| api | web | grafana | prometheus | airflow | db | mlflow |
|---|---|---|---|---|---|---|
| FastAPI | Flask | Grafana | Prometheus | Airflow | SQLite | MLflow |
| :8000 | :8080 | :3000 | :9090 | :8080 | volume | :5000 |
| REST API<br>Predictions<br>Monitoring<br>/health /metrics | Interface<br>utilisateur<br>Jinja2<br>Dashboard | Dashboards<br>Metriques<br>Infinity<br>Prometheus | Scraping<br>metriques<br>/metrics<br>Alertes | DAG fetch<br>DAG predict<br>@daily<br>Logs UI | weather.db<br>Predictions<br>Observations<br>Partage | Model<br>Registry<br>Experiences<br>Artefacts |

**Reseau Docker : projet_final_network | Volume SQLite partage | .env pour variables sensibles | healthcheck sur api**

# Orchestration Airflow & Deploiement Continu (GitHub Actions)

## Pipeline Airflow (schedule: @daily — 01:00 UTC)

| [1] fetch_weather | [2] validate_data | [3] run_predictions | [4] store_results | [5] trigger_report |
|---|---|---|---|---|
| GET Open-Meteo<br>72h previsions | Check non-null<br>Qualite donnees | LSTM inference<br>24 pas x 3h | INSERT SQLite<br>weather.db | Genere rapport<br>monitoring |

Retry: 3 fois | SLA: 30min | Logs: Airflow UI :8080 | Trigger: manuel ou schedule

## Pipeline CI/CD — GitHub Actions (push sur main)

| Step 1 | Step 2 | Step 3 | Step 4 | Step 5 | Step 6 | Step 7 |
|---|---|---|---|---|---|---|
| git push<br>(trigger) | pip install<br>requirements | pytest<br>16 tests | docker<br>build | docker-compose<br>up -d | GET /health<br>assert 200 | Badge<br>statut |

GIT_COMMIT_ID -> SOFTWARE_VERSION | Secrets dans GitHub Secrets | Rollback auto si health check echoue

## FastAPI — Endpoints principaux :

| GET /health | GET /predictions | GET /monitoring/data | POST /pipeline/fetch | POST /pipeline/predict | GET /metrics |
|---|---|---|---|---|---|

# Demonstration

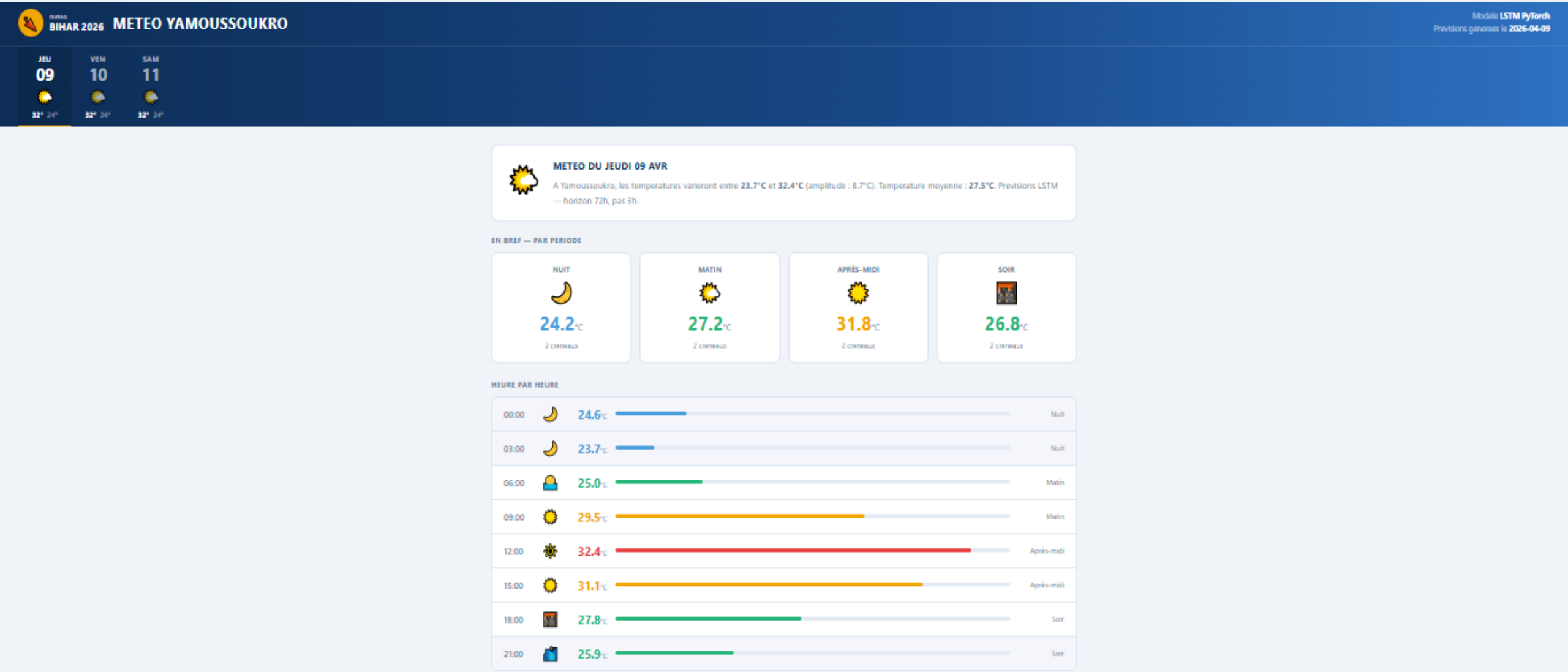
Interface Web · Dashboard Grafana · Tests Automatises

# Interface Web docs

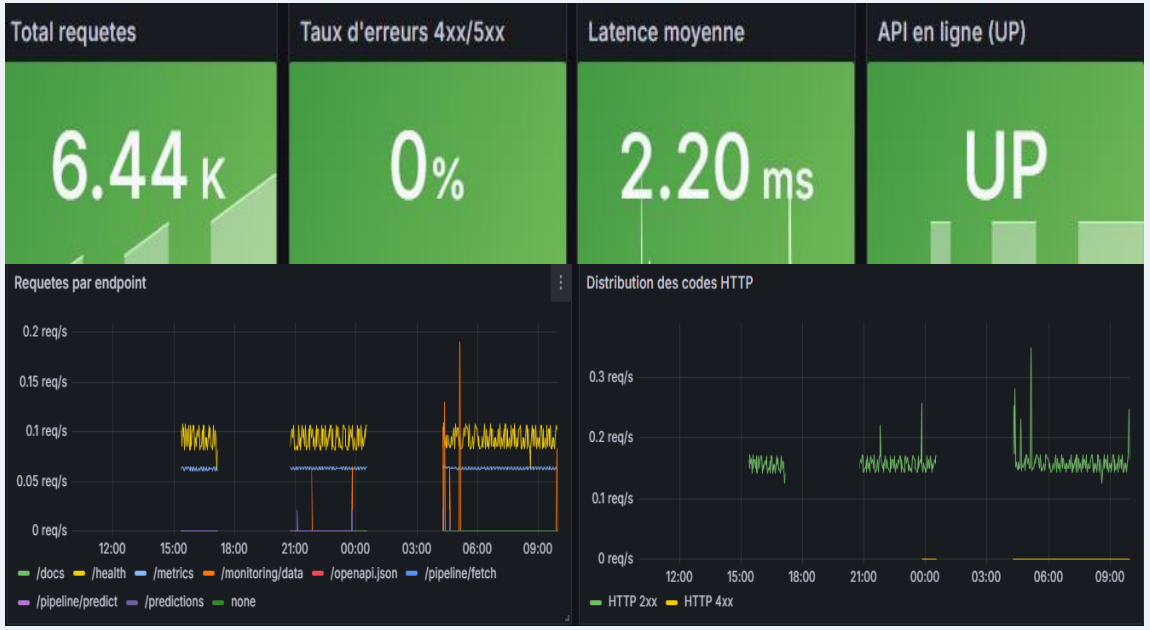
## default

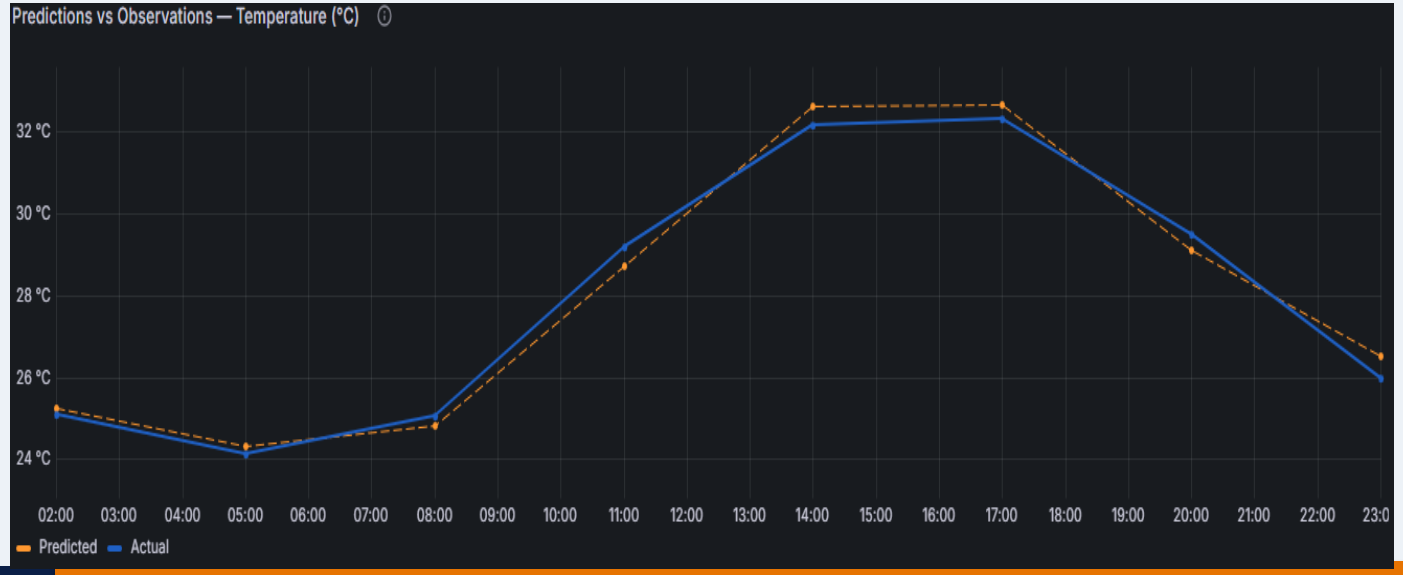
| | | |
|---|---|---|
| GET | /metrics | Metrics |
| GET | /predictions | Prédictions pour une date de génération |
| GET | /predictions/combined | Prédictions + valeurs réelles |
| GET | /version | Version logicielle et version modèle |
| GET | /monitoring/data | Données prédictions vs réel pour Grafana |
| GET | /monitoring/refresh | Recalcul immédiat des métriques Prometheus |
| POST | /pipeline/fetch | Déclenche le fetch des données météo (Airflow) |
| POST | /pipeline/predict | Déclenche la génération des prédictions (Airflow) |

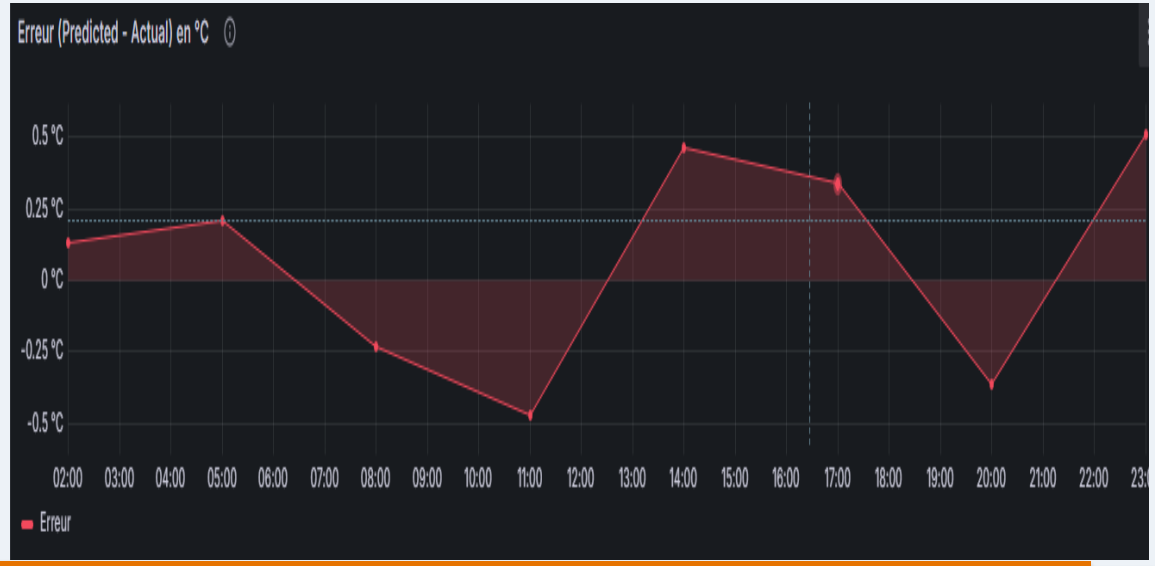
# Interface Web Flask — Dashboard de Prevision Meteorologique



# Dashboard Grafana — Supervision Meteo en Temps Reel



# Dashboard Airflow

Airflow DAGs Cluster Activity Datasets Security Browse Admin Docs 08:13 UTC AA

Triggered weather_forecast_pipeline, it should start any moment now.

## DAGs

All 1 | Active 1 | Paused 0 | Running 0 | Failed 0 | Filter DAGs by tag | Search DAGs | Auto-refresh

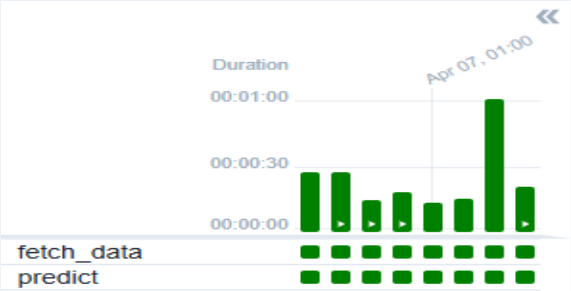
| DAG | Owner | Runs | Schedule | Last Run | Next Run | Recent Tasks | Actio |
|---|---|---|---|---|---|---|---|
| weather_forecast_pipeline (bihar2026, lstm, meteo) | bihar2026 | 7, 1 | 0 1 * * * | 2026-04-09, 08:07:12 | 2026-04-09, 01:00:00 | 1, 1 | |

Airflow DAGs Cluster Activity Datasets Security Browse Admin Docs 08:15 UTC AA

09/04/2026 08:14:59 | All Run Types | All Run States | Clear Filters | Auto-refresh | 25

Press shift + / for Shortcuts

deferred, failed, queued, removed, restarting, running, scheduled, shutdown, skipped, success, up_for_reschedule, up_for_retry, upstream_failed, no_status




DAG
weather_forecast_pipeline

Details | Graph | Gantt | Code | Audit Log | Run Duration | Calendar

| DAG Runs Summary | |
|---|---|
| Total Runs Displayed | 8 |
| Total success | 8 |
| First Run Start | 2026-04-06, 16:38:37 UTC |
| Last Run Start | 2026-04-09, 08:07:13 UTC |
| Max Run Duration | 00:01:00 |

# Dashboard Mlfow

mlflow 2.16.2 Experiments Models GitHub Docs

Experiments

Search Experiments

Default

temperature_forecast

temperature_forecast Provide Feedback Add Description Share

Runs Evaluation Experimental Traces Experimental

user_id = 'DELL' Time created State: Active New run

Datasets Sort: Created Columns Expand rows Group by

| Run Name | Created | Dataset | Duration | Source | Models |
|---|---|---|---|---|---|
| luxuriant-smelt-291 | 7 days ago | weather_test (607e301a) Test , ... +1 | 2.6min | C:\Users... | lstm_temperat |
| big-bear-791 | 7 days ago | weather_test (607e301a) Test , ... +1 | 2.0min | C:\Users... | lstm_temperat |
| treasured-ray-439 | 7 days ago | - | 31ms | C:\Users... | - |
| learned-pug-736 | 7 days ago | weather_test (607e301a) Test , ... +1 | 1.1min | C:\Users... | - |
| bustling-fawn-959 | 7 days ago | - | 2.0min | C:\Users... | lstm_temperat |
| whimsical-stoat-940 | 7 days ago | - | 1.8min | C:\Users... | - |
| likeable-fox-558 | 8 days ago | - | 4.6min | C:\Users... | - |

8 matching runs

mlflow 2.16.2 Experiments Models GitHub Docs

temperature_forecast >

luxuriant-smelt-291 Model registered

Overview Model metrics System metrics Artifacts

| | |
|---|---|
| Created at | 2026-04-02 06:47:32 |
| Created by | DELL |
| Experiment ID | 186270127951935645 |
| Status | Finished |
| Run ID | 63e370023ae94cb6b475eb2c0e0747d1 |
| Duration | 2.6min |
| Datasets used | weather_test (607e301a) Test +2 |
| Tags | Add |
| Source | C:\Users\DELL\Desktop\ESTIA\COURS\MLOps\Projet_final\src\trai |
| Logged models | pytorch |

# Bilan Technique & Perspectives

## Realisations

| | |
|---|---|
| **LSTM 3 couches** | MAE 0.807 C en production (56 pts) |
| **Pipeline ETL** | Airflow @daily — fetch + predict + store |
| **Fusion Multimodale** | Early Fusion F1=0.738 (image+texte) |
| **7 microservices** | Docker Compose orchestre |
| **API REST** | FastAPI : 9 endpoints documentes Swagger |
| **Monitoring** | Grafana + Prometheus temps reel |
| **CI/CD** | GitHub Actions : test -> build -> deploy |
| **Tests auto.** | 16 tests pytest (unit + integration) |

## Perspectives

| | |
|---|---|
| **TFT fine-tune** | Entrainer TFT sur 3 ans de donnees locales |
| **Features meteo** | Ajouter vent, humidite, pression atm. |
| **App mobile** | React Native — Notifications push alertes |
| **Cloud deploy** | AWS/GCP — autoscaling Kubernetes |
| **Entraînement automatique** | Entraînement automatique à travers des derives |
| **A/B testing** | Comparer modeles en production |
| **BERT/GPT** | Tester LLM pour classification texte |
| **Alerting** | MAE > seuil -> alerte Slack/email |

# Conclusion

| 0.807 C | 0.738 | 7 | 16 | 24/24 | 2 ans |
|---|---|---|---|---|---|
| MAE production LSTM 72h | F1 Early Fusion multimodal | Microservices Docker | Tests automatises | Monitoring continu | Donnees entrain. |

**Un systeme MLOps complet :**

Collecte automatique (Airflow) -> Modele LSTM (PyTorch) -> API FastAPI -> Interface Web -> Monitoring Grafana

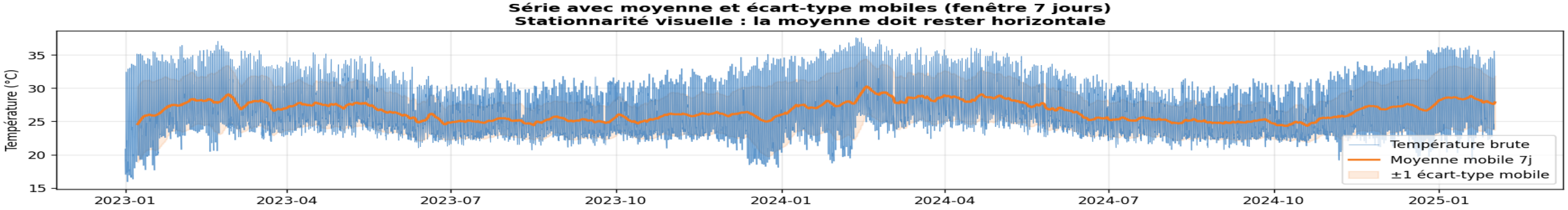
Yamoussoukro, Cote d'Ivoire | Temperature prevue a 72h | MAE < 1 C en production

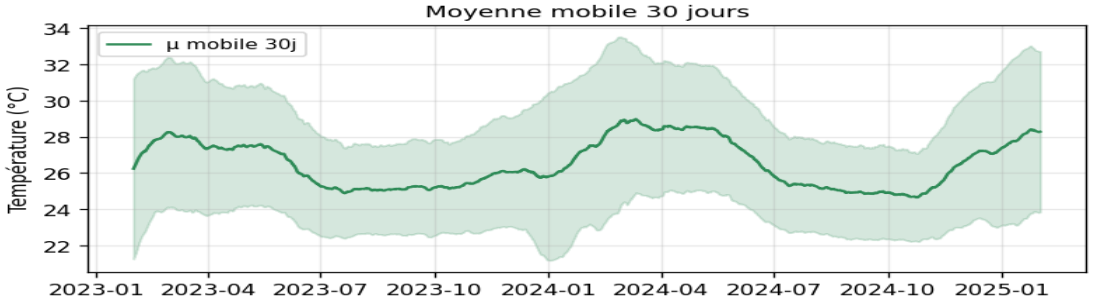
## Merci pour votre attention

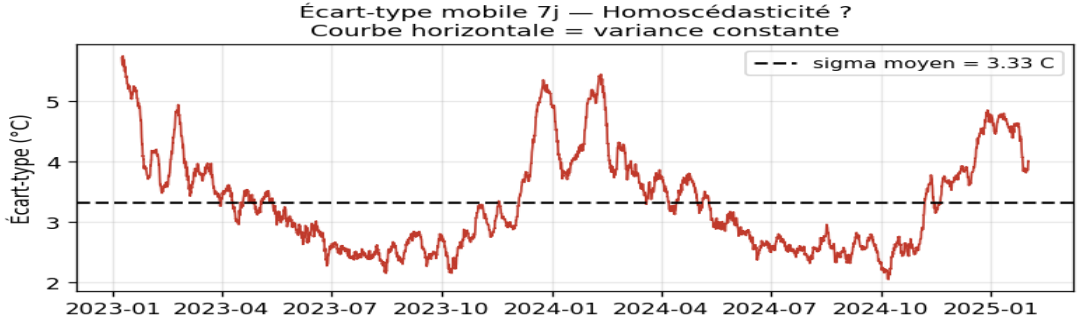
TOURE Abdoul Aziz | Master MLOps ESTIA | Avril 2026 | BIHAR 2026

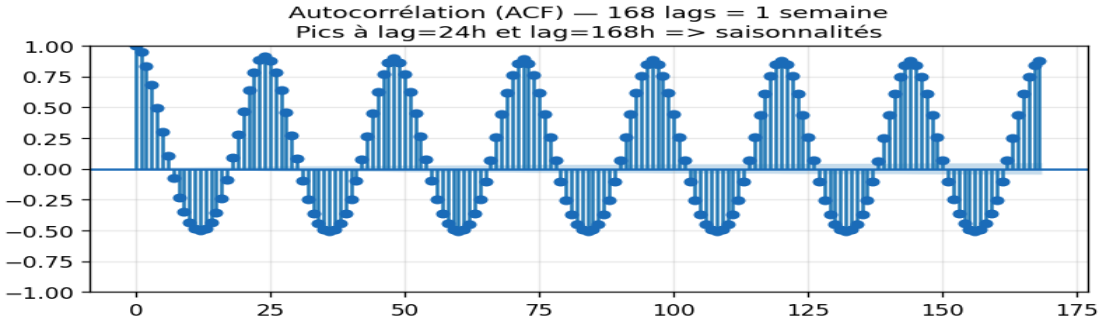
# Annexes

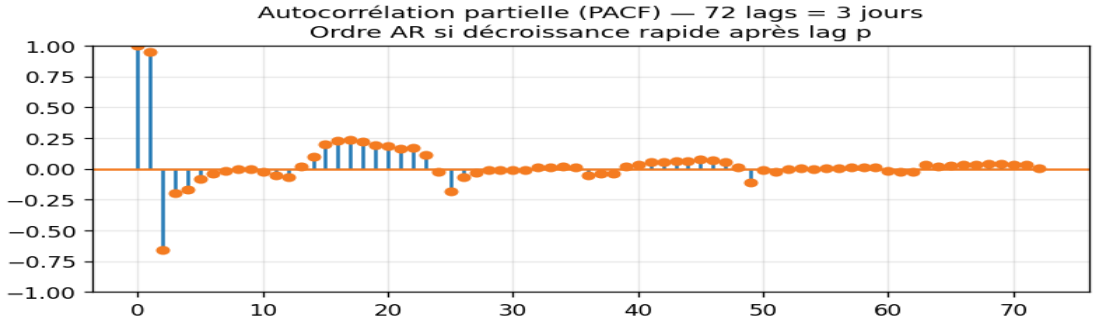
**Analyse de Stationnarité — Température Yamoussoukro (Côte d'Ivoire)**



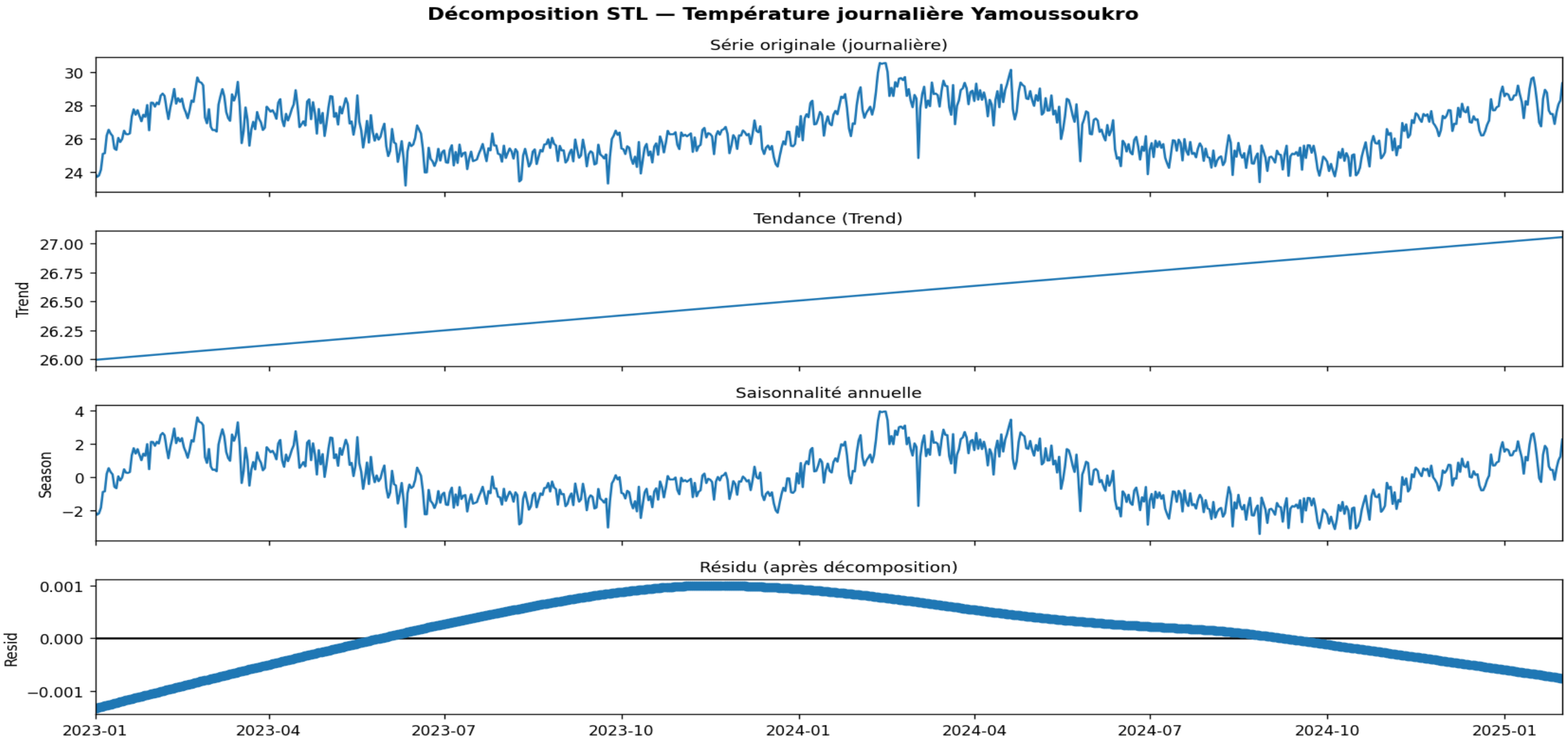



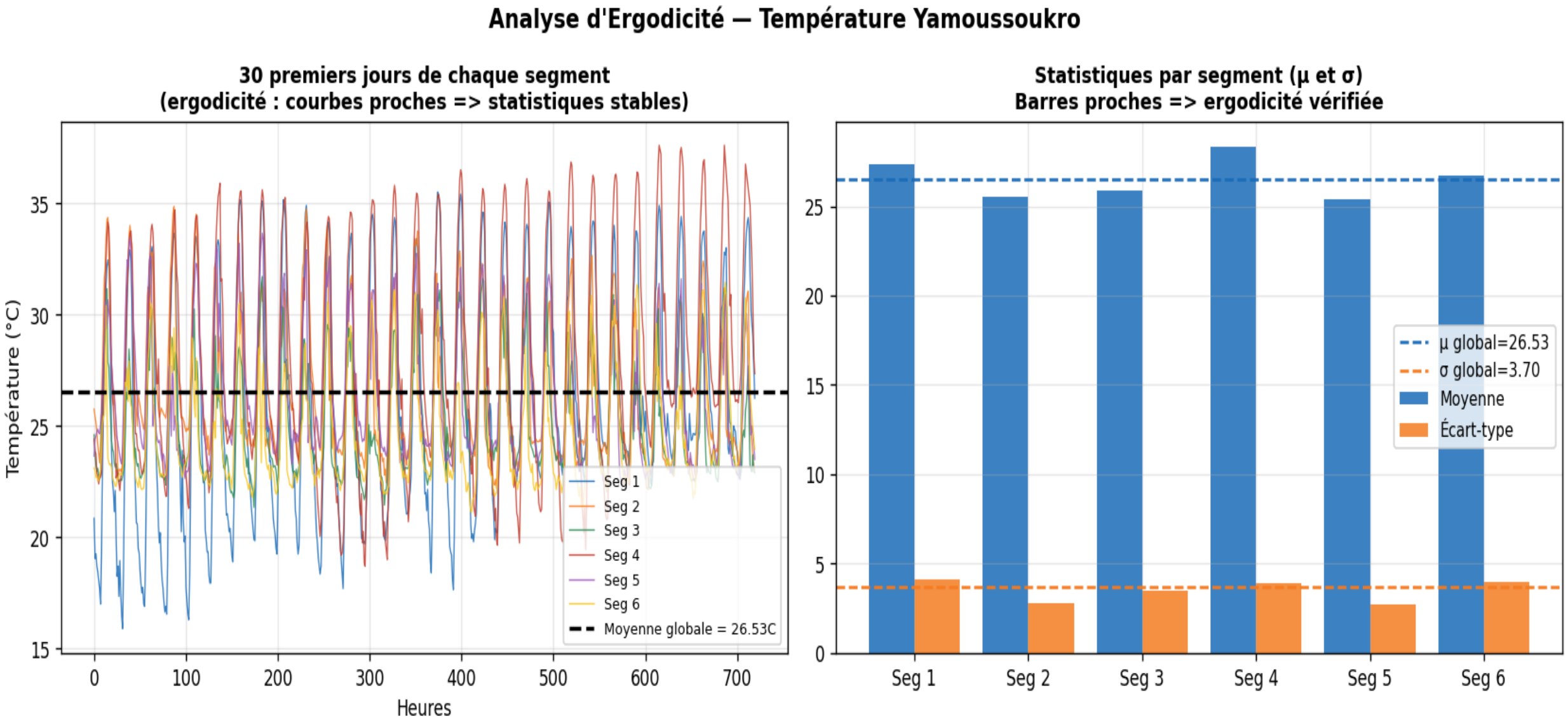



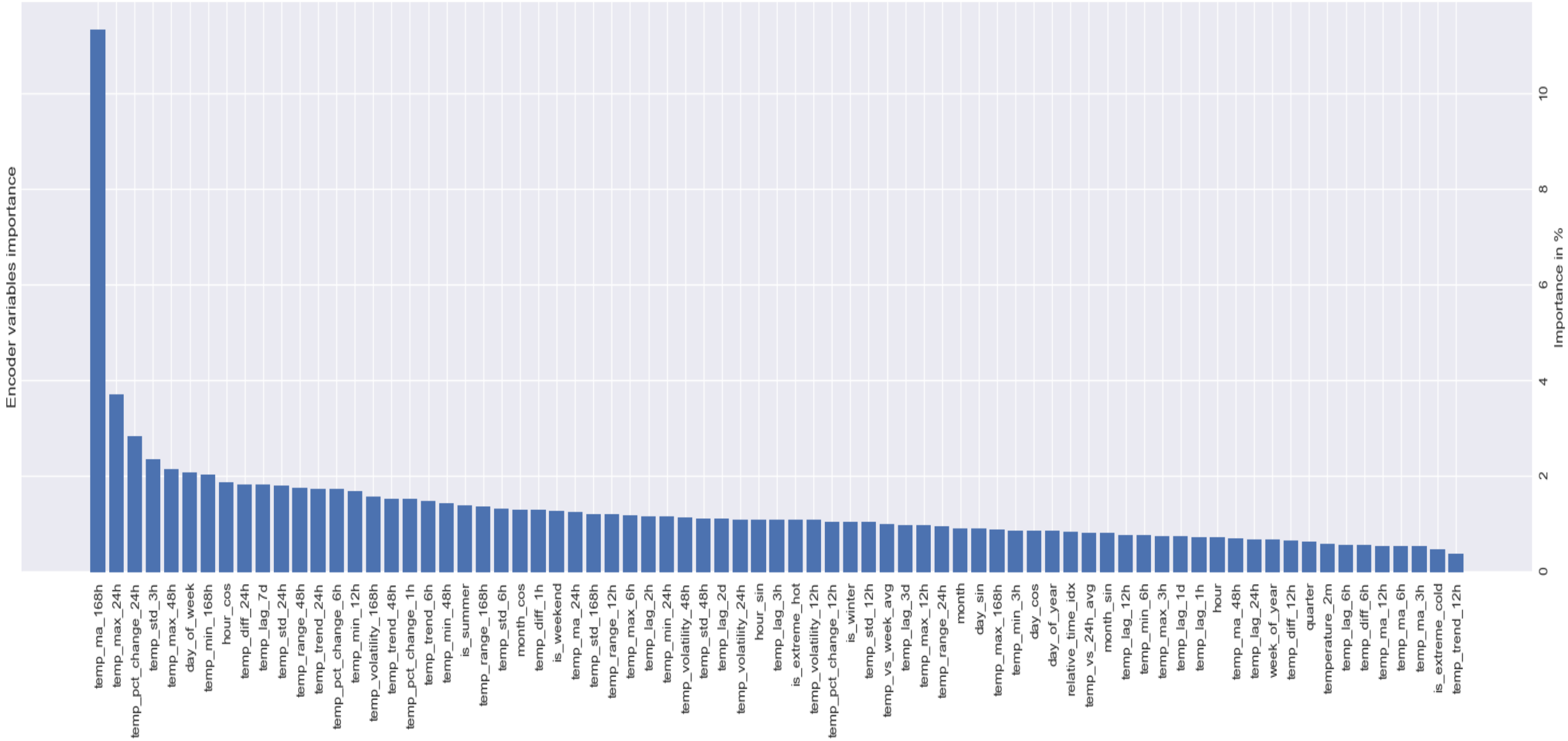
# Annexes


Décomposition STL — Température journalière Yamoussoukro
Série originale (journalière)
30
28
26
24
Tendance (Trend)
Trend
27.00
26.75
26.50
26.25
26.00
Saisonnalité annuelle
Season
4
2
0
−2
Résidu (après décomposition)
Resid
0.001
0.000
−0.001
2023-01
2023-04
2023-07
2023-10
2024-01
2024-04
2024-07
2024-10
2025-01

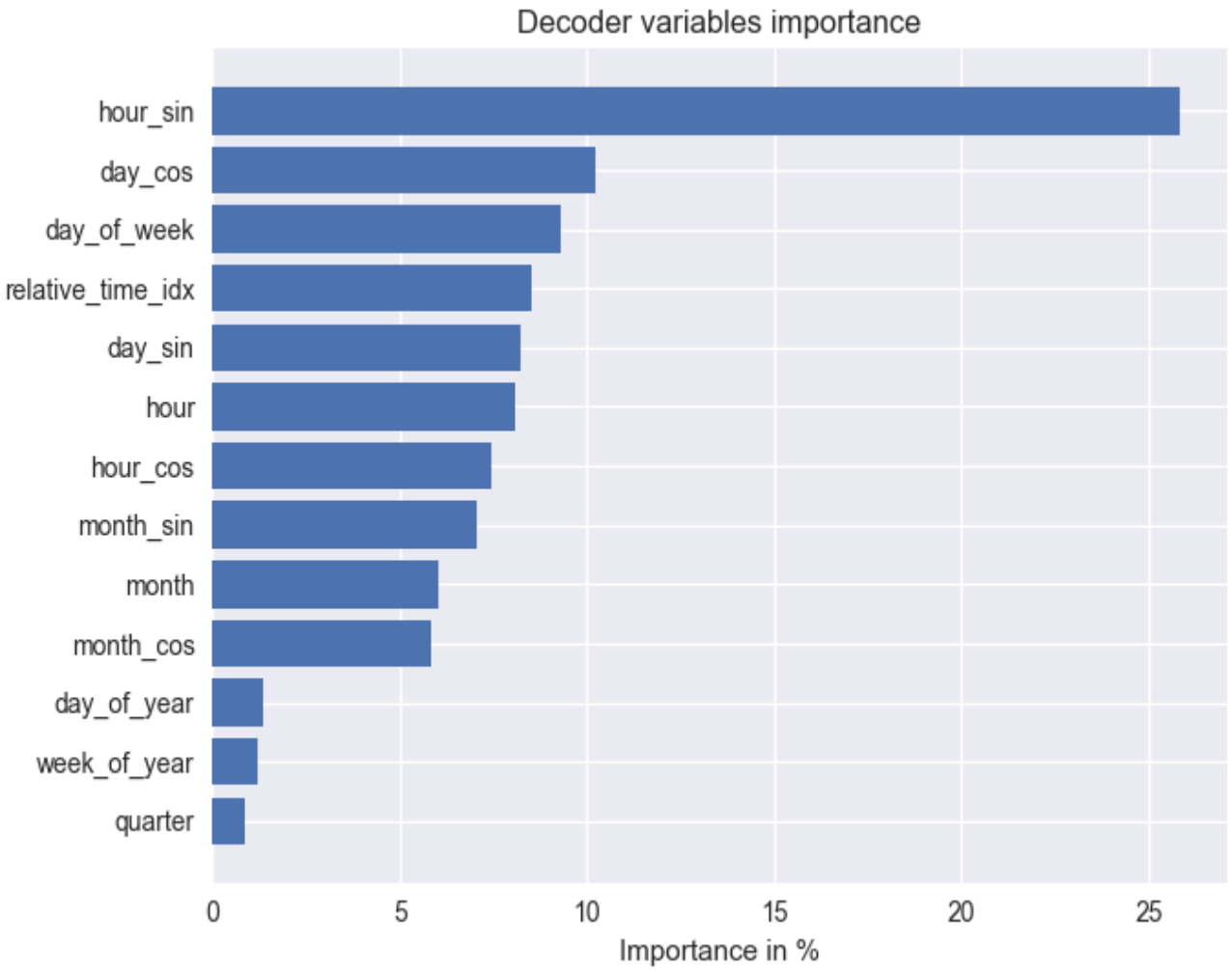
# Annexes

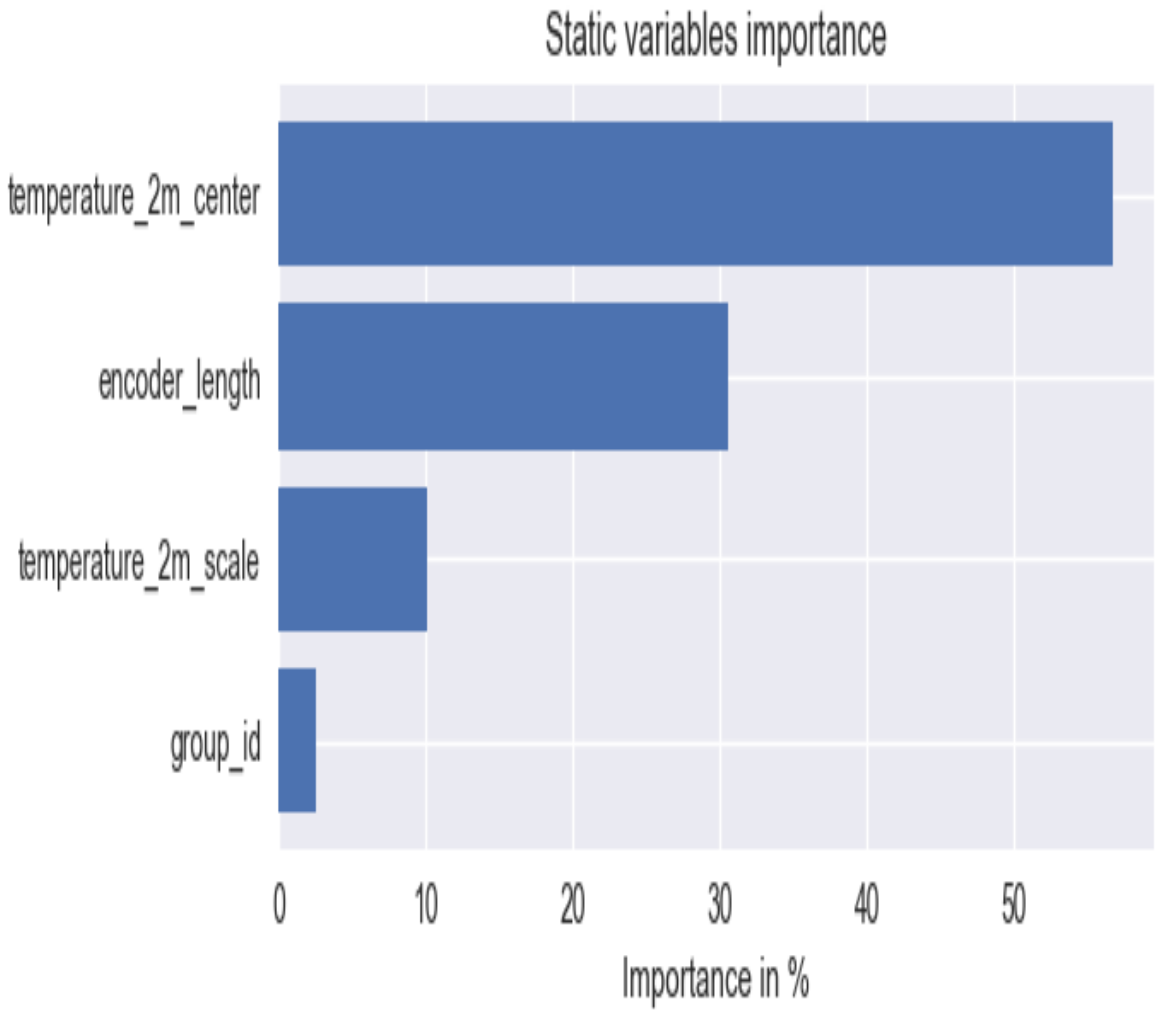

Analyse d'Ergodicité — Température Yamoussoukro
30 premiers jours de chaque segment
(ergodicité : courbes proches => statistiques stables)
Température (°C)
Heures
Seg 1
Seg 2
Seg 3
Seg 4
Seg 5
Seg 6
Moyenne globale = 26.53C
Statistiques par segment (μ et σ)
Barres proches => ergodicité vérifiée
μ global=26.53
σ global=3.70
Moyenne
Écart-type

# Annexes

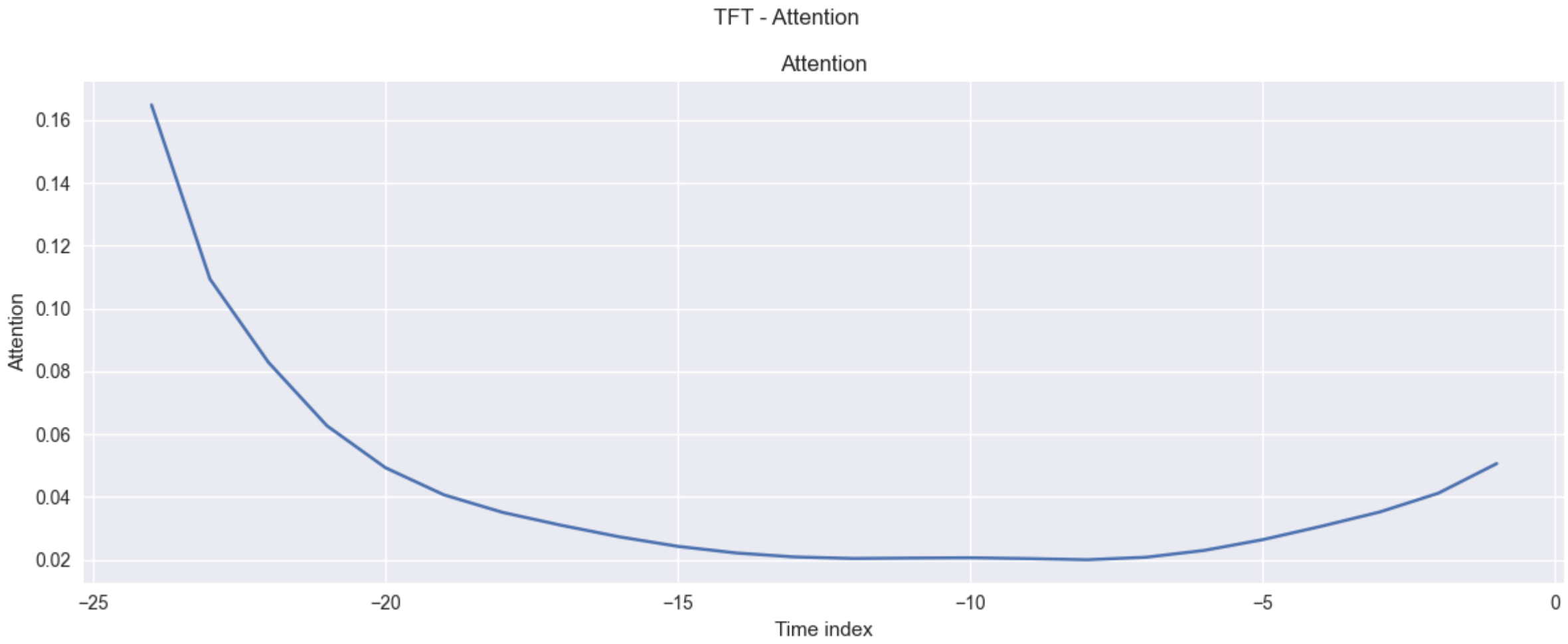

Encoder variables importance
Importance in %
0
2
4
6
8
10
temp_ma_168h
temp_max_24h
temp_pct_change_24h
temp_std_3h
temp_max_48h
day_of_week
temp_min_168h
hour_cos
temp_diff_24h
temp_lag_7d
temp_std_24h
temp_range_48h
temp_trend_24h
temp_pct_change_6h
temp_min_12h
temp_volatility_168h
temp_trend_48h
temp_pct_change_1h
temp_trend_6h
temp_min_48h
is_summer
temp_range_168h
temp_std_6h
month_cos
temp_diff_1h
is_weekend
temp_ma_24h
temp_std_168h
temp_range_12h
temp_max_6h
temp_lag_2h
temp_min_24h
temp_volatility_48h
temp_std_48h
temp_lag_2d
temp_volatility_24h
hour_sin
temp_lag_3h
is_extreme_hot
temp_volatility_12h
temp_pct_change_12h
is_winter
temp_std_12h
temp_vs_week_avg
temp_lag_3d
temp_max_12h
temp_range_24h
month
day_sin
temp_max_168h
temp_min_3h
day_cos
day_of_year
relative_time_idx
temp_vs_24h_avg
month_sin
temp_lag_12h
temp_min_6h
temp_max_3h
temp_lag_1d
temp_lag_1h
hour
temp_ma_48h
temp_lag_24h
week_of_year
temp_diff_12h
quarter
temperature_2m
temp_lag_6h
temp_diff_6h
temp_ma_12h
temp_ma_6h
temp_ma_3h
is_extreme_cold
temp_trend_12h

# Annexes


Decoder variables importance
hour_sin
day_cos
day_of_week
relative_time_idx
day_sin
hour
hour_cos
month_sin
month
month_cos
day_of_year
week_of_year
quarter
0
5
10
15
20
25
Importance in %

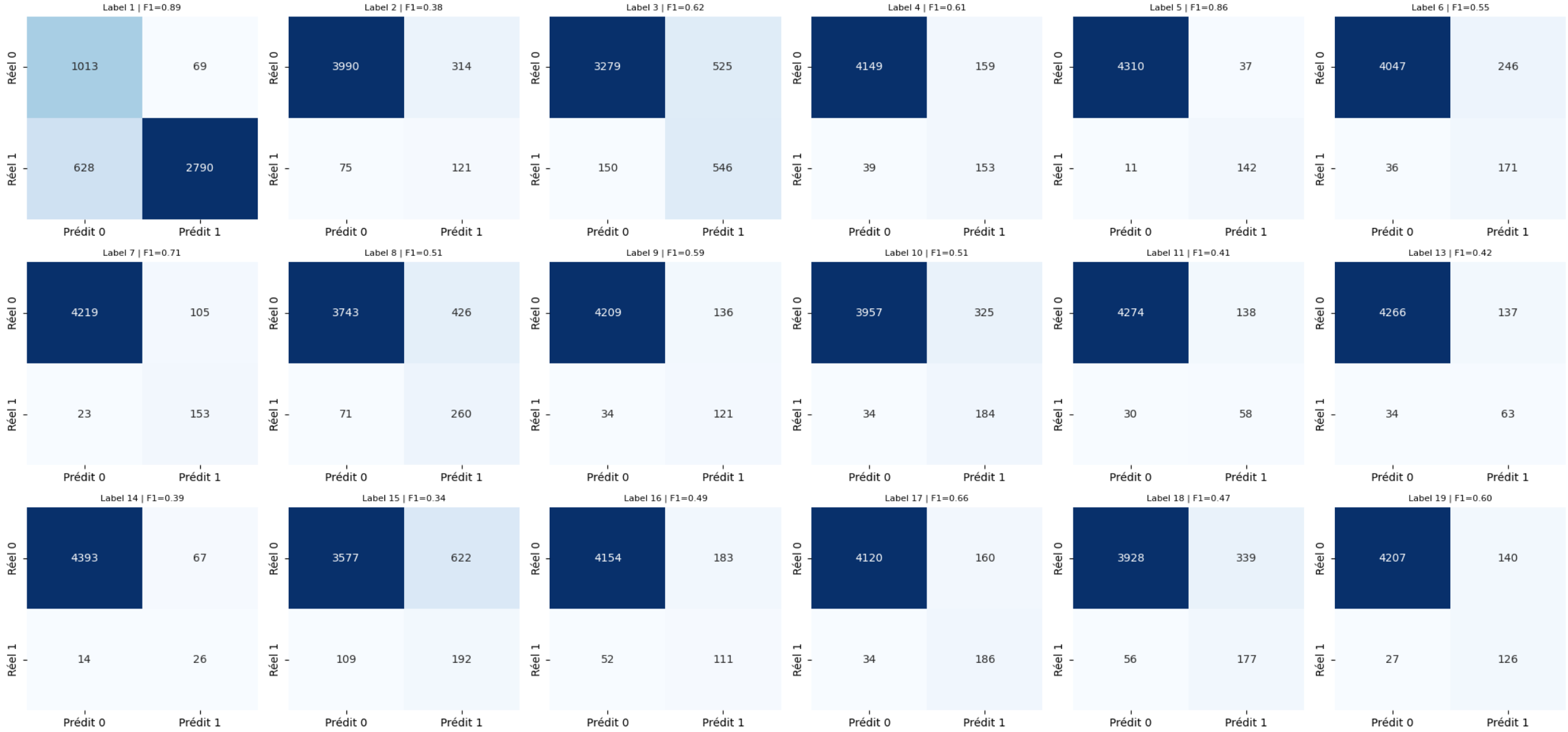

Static variables importance
temperature_2m_center
encoder_length
temperature_2m_scale
group_id
0
10
20
30
40
50
Importance in %

# Annexes

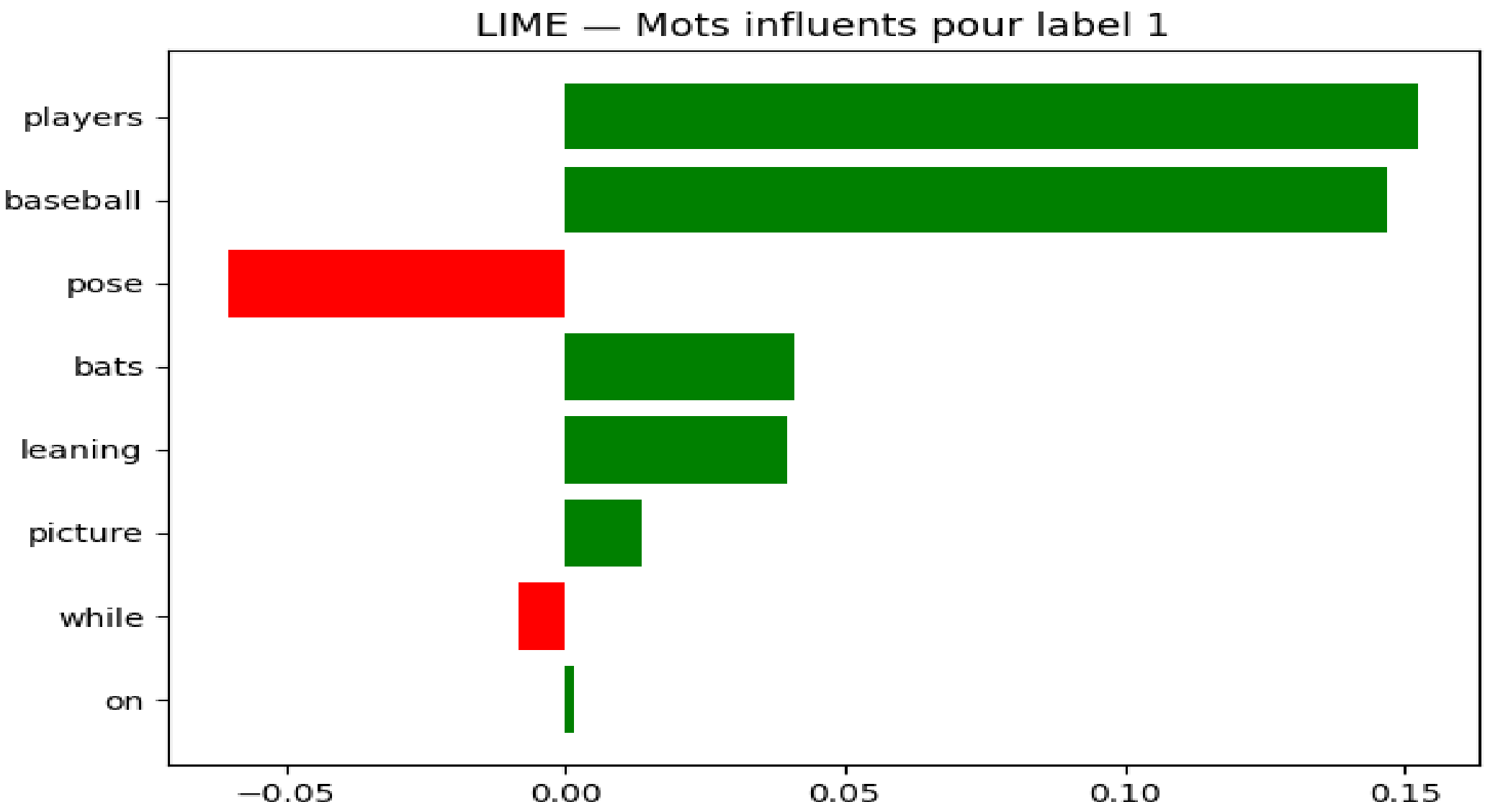

TFT - Attention
Attention
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
Attention
−25
−20
−15
−10
−5
0
Time index

# Annexes

Matrices de confusion par classe (modèle image final)



# Annexes


LIME — Mots influents pour label 1
players
baseball
pose
bats
leaning
picture
while
on
−0.05
0.00
0.05
0.10
0.15

# Annexes

## Résultats comparatifs — MAE par horizon (jeu de test)

| Horizon | Prophet | LSTM | TFT (q50) | Meilleur |
|---|---|---|---|---|
| +3h | 1.3770 | **0.0434** | 1.9465 | LSTM |
| +24h | 1.4100 | **0.0506** | 3.4261 | LSTM |
| +48h | 1.4137 | **0.0537** | 0.4167 | LSTM |
| +72h | 1.4200 | **0.0503** | 1.0303 | LSTM |
| **Moyenne** | 1.4107 | **0.0516** | 1.5763 | **LSTM** |

| Horizon | Prophet RMSE | LSTM RMSE | TFT RMSE |
|---|---|---|---|
| +3h | 1.8954 | **0.0551** | 1.9465 |
| +24h | 1.8547 | **0.0633** | 3.4261 |
| +48h | 1.8556 | **0.0679** | 0.4167 |
| +72h | 1.8605 | **0.0644** | 1.0303 |
| **Moyenne** | 1.8695 | **0.0652** | 1.8644 |